

Northwind Analytics Challenge

MySQL · Aggregate Functions · CSV Export · Charts · Data Storytelling

Course: Data Analytics Level: Intermediate Tool: MySQL + Excel Duration: 2–3 Hours

1. Exercise Overview

In this exercise you will practice writing MySQL aggregate queries against the classic Northwind sample database. You will export each result set to CSV, load it into Excel, build charts and pivot tables, and finally craft a data-driven business narrative from your findings.

Learning Objectives

- Write MySQL queries using aggregate functions: SUM, COUNT, AVG, MIN, MAX
- Apply GROUP BY, HAVING, and ORDER BY to summarise and filter data
- Export result sets from MySQL Workbench to CSV format
- Load CSV data into Excel and create pivot tables and charts
- Practice data storytelling: turning numbers into business insights using the What / So What / Now What framework

Database Context

The Northwind database models a fictional food-import/export company. The key tables for this exercise are:

Table	Description
orders	One row per order — OrderID, CustomerID, EmployeeID, OrderDate, ShipCountry, Freight
`order details`	Line items — OrderID, ProductID, UnitPrice, Quantity, Discount ⚠ Note: table name includes a space — wrap in backticks
products	Product catalogue — ProductID, ProductName, CategoryID, UnitPrice
categories	8 product categories — CategoryID, CategoryName
employees	9 sales staff — EmployeeID, FirstName, LastName, Title
customers	Corporate customers — CustomerID, CompanyName, Country

⚠ Important — Table Name with a Space

The line-items table is named ``order details`` (with a space). In MySQL you must wrap it in backtick characters: ``order details``. Omitting the backticks causes Error 1146: Table does not exist.

2. Setup Instructions

Step 1 —Run the Northwind Database

1. Open MySQL Workbench and connect to your local server
2. Add USE northwind;
3. Run: File > New SQL Script
4. Select Query – Start with Query 1 from StoryTellingEX1.sql file > Execute
5. Verify with:

```
USE northwind;  
SHOW TABLES;
```

Step 2 — Exporting Results to CSV

6. Run each query task 1, 2, 3, 4, and 5 (Ctrl+Enter or click the lightning bolt)
7. In the result grid, click the Export icon or go to File > Export > Export Resultset to CSV
8. Name the file Q1_results.csv (adjust number per query)
9. Ensure encoding is UTF-8 and separator is a comma (,)
10. Open in Excel: Data tab > Get Data > From File > From Text/CSV

3. Data Storytelling Framework

Before writing SQL, understand how your output will tell a story. Great data analysis does not stop at numbers — it answers three questions for every finding:

What?	So What?	Now What?
Describe the pattern or number you observe in the chart.	Explain why this finding matters for the business.	Recommend a concrete action based on the finding.
Example: "Beverages generated \$267K — 21% of all revenue."	"This is the highest-margin category. Over-reliance on one category is a risk."	"Negotiate volume discounts with top Beverages suppliers before Q3."

Tip

For every chart you build, write at least one What / So What / Now What statement before moving to the next query. This keeps your analysis focused and makes the executive summary much easier to write.

4. SQL Query Tasks

Complete all five queries below. Write each query in MySQL Workbench, verify the results, export to CSV, build the corresponding chart in Excel, then answer the storytelling questions.

Query 1 · Revenue by Product Category

Business Question	Which product categories generate the most revenue? Understanding category performance helps buyers prioritise stock and negotiate supplier contracts.
Hint	JOIN path: categories > products > `order details` > orders. Revenue formula: Quantity × UnitPrice × (1 – Discount)
Expected Columns	CategoryName Total_Revenue Total_Orders Avg_Order_Value Total_Units_Sold
Chart Type	Horizontal Bar Chart: CategoryName (Y-axis) vs Total_Revenue (X-axis). Sort bars by value descending. Add data labels showing dollar amounts.

SQL Query 1: available in the StoryTellingEX1 sql file

```

1  -- =====
2  -- Northwind Analytics Challenge
3  -- MySQL · Aggregate Functions · Data Storytelling
4  -- Date: April 24, 2026
5  -- Deodat P. Sharma
6  -- =====
7
8  • USE northwind;
9
10 -- NOTE: The line-items table is named "order details" (with a space).
11 --      It must always be wrapped in backticks: `order details`
12
13 -- =====
14 -- Query 1 – Revenue by Product Category
15 -- Business Question: Which product categories generate the most revenue?
16 -- Chart: Horizontal Bar Chart – CategoryName vs Total_Revenue
17 -- Export: Q1_results.csv
18 -- =====
19 • SELECT
20     c.CategoryName,
21     SUM(od.Quantity * od.UnitPrice * (1 - od.Discount)) AS Total_Revenue,
22     COUNT(DISTINCT o.OrderID) AS Total_Orders,
23     AVG(od.Quantity * od.UnitPrice * (1 - od.Discount)) AS Avg_Order_Value,
24     SUM(od.Quantity) AS Total_Units_Sold
25 FROM categories      c
26 JOIN products        p ON p.CategoryID = c.CategoryID
27 JOIN `order details` od ON od.ProductID = p.ProductID
28 JOIN orders          o ON o.OrderID    = od.OrderID
29 GROUP BY c.CategoryName
30 ORDER BY Total_Revenue DESC;
31

```

Export to CSV

Run query in MySQL Workbench → File > Export > Export Results to CSV → Save as Q1_results.csv

Data Storytelling

What? Which category had the highest Total_Revenue? By how much did it lead the second-place category?

So What? What does this dominance imply about Northwind's product strategy or customer preferences?

Now What? Which two categories should the sales team prioritise next quarter, and why?

Bonus: Which category had the lowest Avg_Order_Value? Is low average order size a volume or pricing problem?

Query 2 · Monthly Revenue Trend

Business Question	How does revenue fluctuate month-to-month and year-over-year? Identifying seasonal peaks helps the operations team plan inventory and staffing levels.
Hint	Use DATE_FORMAT('%b') for readable month labels (Jan, Feb ...). Sort by numeric Year and Month, not by MonthName string.
Expected Columns	Year Month MonthName Total_Revenue Num_Orders
Chart Type	Line Chart: Month (X-axis) vs Total_Revenue (Y-axis). Plot two separate lines — one for 2023 (blue) and one for 2024 (orange). Add a legend and gridlines.

SQL Query 2 available in the StoeyTellingEX1 sql file

```

31  -- =====
32  -- Query 2 – Monthly Revenue Trend
33  -- Business Question: How does revenue fluctuate month-to-month and year-over-year?
34  -- Chart: Line Chart – Month vs Total_Revenue (two lines: 2023 and 2024)
35  -- Export: Q2_results.csv
36  -- =====
37  •  SELECT
38      YEAR(o.OrderDate)          AS Year,
39      MONTH(o.OrderDate)         AS Month,
40      DATE_FORMAT(o.OrderDate, '%b') AS MonthName,
41      SUM(od.Quantity * od.UnitPrice * (1 - od.Discount)) AS Total_Revenue,
42      COUNT(DISTINCT o.OrderID)   AS Num_Orders
43  FROM orders                    o
44  JOIN `order details` od ON od.OrderID = o.OrderID
45  GROUP BY Year, Month, MonthName
46  ORDER BY Year, Month;

```

Export to CSV
Run query in MySQL Workbench → File > Export > Export Results to CSV → Save as Q2_results.csv

Data Storytelling
What? Describe the months with peak revenue. Is there a clear seasonal pattern?
So What? Did overall revenue grow from 2023 to 2024? Estimate the year-over-year % change.
Now What? Were there months where revenue dipped significantly? What operational response would you recommend?

Bonus: Does order count (Num_Orders) track closely with revenue, or do some months show high orders but low revenue?

Query 3 · Top Sales Employees	
Business Question	Which employees drive the most revenue? Identifying high performers informs bonus structures and helps management understand workload distribution.
Hint	Use CONCAT() to merge FirstName and LastName. GROUP BY EmployeeID as well as names to avoid grouping errors if two employees share a name.
Expected Columns	Employee Total_Revenue Orders_Handled Avg_Revenue_Per_Order Min_Order_Value Max_Order_Value
Chart Type	Clustered Column Chart: Employee (X-axis) with two bars per employee — Total_Revenue and Orders_Handled (dual Y-axis). Sort by Total_Revenue descending.

SQL Query 3 available in the StoryTellingEX1 sql file

```

48  -- =====
49  -- Query 3 – Top Sales Employees
50  -- Business Question: Which employees drive the most revenue?
51  -- Chart: Clustered Column Chart – Employee vs Total_Revenue & Orders_Handled
52  -- Export: Q3_results.csv
53  -- =====
54  •  SELECT
55      CONCAT(e.FirstName, ' ', e.LastName)           AS Employee,
56      SUM(od.Quantity * od.UnitPrice * (1 - od.Discount)) AS Total_Revenue,
57      COUNT(DISTINCT o.OrderID)                       AS Orders_Handled,
58      AVG(od.Quantity * od.UnitPrice * (1 - od.Discount)) AS Avg_Revenue_Per_Order,
59      MIN(od.Quantity * od.UnitPrice * (1 - od.Discount)) AS Min_Order_Value,
60      MAX(od.Quantity * od.UnitPrice * (1 - od.Discount)) AS Max_Order_Value
61  FROM employees      e
62  JOIN orders          o ON o.EmployeeID = e.EmployeeID
63  JOIN `order details` od ON od.OrderID   = o.OrderID
64  GROUP BY e.EmployeeID, e.FirstName, e.LastName
65  ORDER BY Total_Revenue DESC;
66

```

Export to CSV

Run query in MySQL Workbench → File > Export > Export Results to CSV → Save as Q3_results.csv

Data Storytelling

What? Who is the top-performing employee by Total_Revenue? Is a higher Orders_Handled count correlated with higher revenue?

So What? How large is the revenue gap between the #1 and last-ranked employee? What are the implications?

Now What? Which employees might benefit from coaching or workload rebalancing? How would you structure a bonus plan?

Bonus: Compare Avg_Revenue_Per_Order across employees. Does a lower order count always mean lower performance?

Query 4 · Revenue by Ship Country

Business Question	Which markets contribute the most revenue? Geographic analysis helps the sales team decide where to focus expansion efforts and identify high-freight routes to optimise.
Hint	Use HAVING (not WHERE) to filter on aggregated results. LIMIT 10 returns only the top 10 countries by revenue.
Expected Columns	ShipCountry Total_Revenue Num_Orders Avg_Freight Total_Freight_Cost

Chart Type	Pie Chart: One slice per ShipCountry, sized by Total_Revenue. Show percentage labels and country names. Include a legend.
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SQL Query 4 available in the StoryTellingEX1 sql file

```
67      -- =====
68      -- Query 4 – Revenue by Ship Country (Top 10)
69      -- Business Question: Which markets contribute the most revenue?
70      -- Chart: Pie Chart – ShipCountry slices by Total_Revenue
71      -- Export: Q4_results.csv
72      -- =====
73      •  SELECT
74          o.ShipCountry,
75          SUM(od.Quantity * od.UnitPrice * (1 - od.Discount)) AS Total_Revenue,
76          COUNT(DISTINCT o.OrderID) AS Num_Orders,
77          AVG(o.Freight) AS Avg_Freight,
78          SUM(o.Freight) AS Total_Freight_Cost
79      FROM orders o
80      JOIN `order details` od ON od.OrderID = o.OrderID
81      GROUP BY o.ShipCountry
82      HAVING Total_Revenue > 10000
83      ORDER BY Total_Revenue DESC
84      LIMIT 10;
```

Export to CSV
Run query in MySQL Workbench → File > Export > Export Results to CSV → Save as Q4_results.csv

Data Storytelling
What? Which country generates the most revenue? What share of total revenue does it represent?
So What? Are there markets with high Num_Orders but low revenue, suggesting small basket sizes?
Now What? Which countries have a high Avg_Freight relative to revenue — potentially squeezing margins?

Bonus: Are there obvious target markets for expansion based on geographic proximity to top performers?

Query 5 · Category Revenue — Year-over-Year Pivot	
Business Question	How has each category's revenue changed from 2023 to 2024? Year-over-year comparison reveals which categories are growing, stable, or declining.
Hint	Use conditional SUM with CASE WHEN to pivot years into columns. NULLIF(x, 0) prevents division-by-zero errors in the growth % formula.

Expected Columns	CategoryName Revenue_2023 Revenue_2024 Total_Revenue YoY_Growth_Pct
Chart Type	Pivot Chart — Clustered Column: CategoryName (X-axis), two bars per category for 2023 (navy) and 2024 (teal). Add a data table below showing YoY_Growth_Pct values.

SQL Query 5 available in the StoryTellingEX1 sql file

```
86      -- =====
87      -- Query 5 – Category Revenue Year-over-Year Pivot
88      -- Business Question: How has each category's revenue changed from 2023 to 2024?
89      -- Chart: Pivot Chart – Clustered Column (2023 navy, 2024 teal) with data table
90      -- Export: Q5_results.csv
91      -- =====
92      • SELECT
93          c.CategoryName,
94          SUM(CASE WHEN YEAR(o.OrderDate) = 2023
95              THEN od.Quantity * od.UnitPrice * (1 - od.Discount)
96              ELSE 0 END) AS Revenue_2023,
97          SUM(CASE WHEN YEAR(o.OrderDate) = 2024
98              THEN od.Quantity * od.UnitPrice * (1 - od.Discount)
99              ELSE 0 END) AS Revenue_2024,
100         SUM(od.Quantity * od.UnitPrice * (1 - od.Discount)) AS Total_Revenue,
101         ROUND(
102             (SUM(CASE WHEN YEAR(o.OrderDate) = 2024
103                 THEN od.Quantity * od.UnitPrice * (1 - od.Discount)
104                 ELSE 0 END) -
105             SUM(CASE WHEN YEAR(o.OrderDate) = 2023
106                 THEN od.Quantity * od.UnitPrice * (1 - od.Discount)
107                 ELSE 0 END)) /
108             NULLIF(SUM(CASE WHEN YEAR(o.OrderDate) = 2023
109                 THEN od.Quantity * od.UnitPrice * (1 - od.Discount)
110                 ELSE 0 END), 0) * 100,
111             1) AS YoY_Growth_Pct
112     FROM categories      c
113     JOIN products        p ON p.CategoryID = c.CategoryID
114     JOIN `order details` od ON od.ProductID = p.ProductID
115     JOIN orders          o ON o.OrderID    = od.OrderID
116     GROUP BY c.CategoryName
117     ORDER BY Total Revenue DESC;
```

Export to CSV

Run query in MySQL Workbench → File > Export > Export Results to CSV → Save as Q5_results.csv

Data Storytelling

What? Which categories posted positive YoY_Growth_Pct? Which declined?

So What? A declining high-revenue category is a strategic risk. A growing small category may signal an emerging opportunity.

Now What? Which categories warrant increased inventory investment for 2025? Which need a pricing or promotion review?

Bonus: If one category grew in revenue but declined in units (from Q1), what does that imply about pricing power?

5. Excel: Pivot Table & Chart Guide

After loading your Q5 CSV into Excel, follow these steps to create a formal Pivot Table and Pivot Chart using the Q5 data.

Building the Pivot Table

11. Select your Q5 data range including headers (CategoryName, Revenue_2023, Revenue_2024, Total_Revenue, YoY_Growth_Pct)
12. Go to Insert tab > PivotTable > New Worksheet
13. Drag CategoryName to the Rows area
14. Drag Revenue_2023 and Revenue_2024 to the Values area (ensure both are set to Sum)
15. Drag YoY_Growth_Pct to the Values area as a third measure
16. Format Value fields: right-click > Value Field Settings > Number Format > Currency (\$) for revenue, Percentage for growth
17. Sort by Total_Revenue: click any revenue cell > Home > Sort & Filter > Sort Largest to Smallest

Building the Pivot Chart

18. Click anywhere inside your pivot table
19. Go to PivotTable Analyze tab > PivotChart
20. Select Clustered Column chart type > OK
21. Right-click the 2023 series > Format Data Series > Fill colour: Navy (#1F3864)
22. Right-click the 2024 series > Format Data Series > Fill colour: Teal (#17A589)
23. Add chart title: "Category Revenue — 2023 vs 2024"
24. Add axis titles: "Product Category" (X-axis) and "Revenue (USD)" (Y-axis)
25. Right-click chart > Add Data Table to display values below the chart

6. Data Storytelling — Your Written Analysis

Once all five charts are built, use this section to write your analysis. Apply the What / So What / Now What framework to every finding. Use specific numbers from your charts — never vague statements like 'sales were good'.

Category Analysis (Q1)

Question	Your Answer
Which category generated the highest revenue, and by what dollar margin did it lead the second-place category?	
Which category underperformed? What could explain the lower sales volume — limited SKUs, pricing, or customer demand?	
Based on the data, which two categories should the sales team prioritise next quarter, and why?	

Monthly Trend Analysis (Q2)

Question	Your Answer
Describe the months with peak revenue. Is there a clear seasonal pattern (e.g., Q4 holiday spike)?	
Did overall revenue grow from 2023 to 2024? Calculate and state the year-over-year percentage change.	
Were there months where revenue dipped significantly? What factors (operational or external) might explain this?	

Employee Performance (Q3)

Question	Your Answer
Who is the top-performing employee by Total_Revenue? Is a higher Orders_Handled count correlated with higher revenue?	
How large is the revenue gap between the #1 and lowest-ranked employee? What are the implications for the business?	
Does a high Avg_Revenue_Per_Order suggest quality accounts, high-value products, or larger order quantities?	

Geographic Analysis (Q4)

Question	Your Answer
Which country generates the most revenue? What percentage of total top-10 revenue does it account for?	
Are there countries with high Avg_Freight relative to Total_Revenue, suggesting thin margins on shipping?	
Which market looks like the best candidate for increased investment, and which might be worth deprioritising?	

Year-over-Year Category Pivot (Q5)

Question	Your Answer
Which categories posted positive YoY_Growth_Pct? Were any categories in decline?	
If a high-revenue category is declining, what strategic risk does that pose for Northwind's business model?	
Which categories should receive increased investment in 2025 based on growth trajectory?	

7. Executive Summary

Write a 4–5 sentence executive summary of Northwind's business performance, drawing on findings from all five queries. Imagine you are presenting to a senior leadership team. Your summary must include:

- At least one specific dollar figure or percentage from your analysis
- A reference to the strongest-performing dimension (category, employee, or country)
- One risk or area of concern identified in the data
- One forward-looking recommendation for leadership

Write your executive summary here...

8. Completion Checklist

Verify every item below is complete..

✓	Item	Description
[]	SQL	All 5 MySQL queries run without errors in MySQL Workbench
[]	CSV	5 CSV files exported: Q1_results.csv through Q5_results.csv
[]	Charts	Bar (Q1), Line (Q2), Column (Q3), Pie (Q4) — all formatted with titles and axis labels
[]	Pivot Table	Q5 pivot table created in Excel with Revenue_2023 and Revenue_2024 as value fields
[]	Pivot Chart	Q5 pivot chart built as a clustered column chart with data table below
[]	Storytelling	All What/So What/Now What questions answered with chart evidence
[]	Executive Summary	4–5 sentence summary completed, citing specific numbers

Appendix — Aggregate Functions Quick Reference

Function	Purpose	Example
SUM()	Adds all values in a column	SUM(od.Quantity * od.UnitPrice)
COUNT()	Counts rows or distinct values	COUNT(DISTINCT o.OrderID)
AVG()	Calculates the arithmetic mean	AVG(od.UnitPrice)
MIN()	Returns the smallest value	MIN(od.Quantity * od.UnitPrice)
MAX()	Returns the largest value	MAX(od.Quantity * od.UnitPrice)
GROUP BY	Groups rows sharing a column value	GROUP BY c.CategoryName
HAVING	Filters groups (like WHERE for aggregates)	HAVING SUM(...) > 10000
ORDER BY	Sorts the result set	ORDER BY Total_Revenue DESC
NULLIF()	Returns NULL if two values are equal (avoids /0)	NULLIF(revenue_2023, 0)
CASE WHEN	Conditional logic inside aggregates	SUM(CASE WHEN YEAR = 2024 THEN val ELSE 0 END)

Common Troubleshooting
Error 1146 (Table does not exist): Check SHOW TABLES — the line-items table is `order details` with a space, requiring backticks. Error 1052 (Column ambiguous): Prefix column names with the table alias (e.g., o.OrderID instead of OrderID). Error 1064 (Syntax): Check for missing commas between SELECT columns or missing JOIN ON clauses.